

Erfan Jaripour

Tehran, Iran •

Email: erfanjaripour@gmail.com •

Personal Website: [erfanjaripour/github.io](https://erfanjaripour.github.io) •

GitHub: github.com/erfanjaripour •

ORCID: [0009-0009-3229-4721](https://orcid.org/0009-0009-3229-4721) •

OSF: osf.io/j2ws4

RESEARCH INTERESTS: Computational Cognitive Science, Working Memory, Decision-Making, Cognitive Modeling, Reproducible Research

Education

Bachelor of Psychology, Islamic Azad University, Tehran, Iran

2021–2025

GPA: 19.55/20 (equivalent approximately 3.9/4.0)

THESIS: *Relationship Between Psychological Disorders and Defense Mechanisms in Adolescents*

SUPERVISOR: Dr. Hamid Reza Vatankhah

Selected Coursework: Descriptive Statistics; Inferential Statistics; Cognitive Psychology; Experimental Psychology; Psychometrics; Neurological Sciences; Research in Psychology

Additional Quantitative Preparation: Linear Algebra; Calculus; Probability Theory; Scientific Programming with Python and R; Data Analysis and Visualization

Research Experience

Independent Computational Research Projects

2025–Present

Stroop Interference: A Behavioral and Computational Analysis

June 2025

- Analyzed trial-level Stroop task data from 81 participants using Python.
- Cleaned and validated behavioral datasets, including participant–session structure resolution and quality-control procedures.
- Quantified Stroop interference through descriptive statistics, paired-samples inference, effect-size estimation, and robustness analyses.
- Examined individual differences in interference magnitude across participants.
- Implemented a simple computational simulation to reproduce observed condition-level behavioral patterns.
- Produced a fully reproducible research workflow with documented analyses, visualizations, and manuscript preparation.

GITHUB REPOSITORY: github.com/erfanjaripour/stroop-interference

Computational Cognitive Science Training

Mar 2025

- Structured self-directed training in Python-based data analysis for cognitive science and behavioral research.
- Developed practical skills in NumPy, Pandas, Matplotlib/Seaborn, and inferential statistics.
- Completed mini-projects after each module, including data simulation, cleaning, visualization, and hypothesis testing.
- Built reproducible analysis workflows with clear notebook organization and documented code.
- Applied statistical methods such as descriptive analysis, regression, and basic model interpretation to research-oriented datasets.

GITHUB REPOSITORY: github.com/erfanjaripour/computational-cognitive-science-training

Publications & Preprints

Erfan Jaripour (2026). *Stroop Interference in Reaction Time and Accuracy: A Behavioral and Computational Analysis*.

PsyArXiv preprint: osf.io/j2ws4/

Technical Skills

PROGRAMMING: Python, R

SCIENTIFIC LIBRARIES: pandas, NumPy, scipy, matplotlib, seaborn, statsmodels

STATISTICAL METHODS: descriptive statistics, inferential statistics, hypothesis testing, ANOVA, regression, repeated-measures designs, effect-size estimation, confidence intervals, exploratory data analysis

RESEARCH WORKFLOW: Git, GitHub, Jupyter Notebooks, Markdown, LaTeX, reproducible research

Honors & Awards

- Tuition Waiver Scholarship, Islamic Azad University, 2024
- Top-Ranked Psychology Student, Islamic Azad University, 2021–2025
- Outstanding Academic Performance Award, Islamic Azad University, 2023

Languages

Persian (Native)

English (Professional Working Proficiency)